

How does a hypha grow? The biophysics of pressurized growth in fungi

Roger R. Lew

Abstract | The mechanisms underlying the growth of fungal hyphae are rooted in the physical property of cell pressure. Internal hydrostatic pressure (turgor) is one of the major forces driving the localized expansion at the hyphal tip which causes the characteristic filamentous shape of the hypha. Calcium gradients regulate tip growth, and secretory vesicles that contribute to this process are actively transported to the growing tip by molecular motors that move along cytoskeletal structures. Turgor is controlled by an osmotic mitogen-activated protein kinase cascade that causes *de novo* synthesis of osmolytes and uptake of ions from the external medium. However, as discussed in this Review, turgor and pressure have additional roles in hyphal growth, such as causing the mass flow of cytoplasm from the basal mycelial network towards the expanding hyphal tips at the colony edge.

At the edge of a fungal colony, leading hyphae grow into new territory in search of food. Behind the colony edge, the hyphae interconnect to form a three-dimensional network that is optimized to extract nutrients from the surrounding medium in order to fuel continued exploration (FIG. 1). Colony growth can be fast (about $10\text{--}100\ \mu\text{m min}^{-1}$, depending on the organism, nutrient availability and temperature) and involves the continuous synthesis of all the cellular constituents that are necessary for rapid cell expansion. A major driving force for cell expansion is pressure.

Pressure is a thermodynamic state property that affects the life of all organisms. In cells that lack a cell wall, excessive pressure can result in cell lysis and death. In cells that do have a wall (most bacteria, algae, fungi and plants), an internal hydrostatic pressure (turgor) provides both mechanical support for free-standing structures and a force that drives cellular expansion, substrate penetration and other processes. Extreme examples from fungi are the projectile release of spores at $>100,000\ \times g$ (g is the acceleration due to gravity at the Earth's surface) in ascomycetes^{1,2} and zygomycetes³, and the application of force that is sufficient for pathogenic fungi to penetrate and ramify throughout host tissue³. The turgor is created by osmosis that occurs in response to the concentrations of osmotically active substances (osmolytes) being higher within the cell than in the surrounding milieu. During cell growth (expansion and division), remodeling of the cell wall is required to increase the cellular volume. Much of our understanding of the mechanisms underlying cellular expansion has been revealed from

studies of tip-growing cells, which expand by tubular extension. Examples of tip-growing model systems are found in many phylogenetic clades and include algae, fungi, fungal allies, and the pollen tubes and root hairs of plants. Tip growth is orthogonal, exhibiting a gradient of expansion that is maximal at the tip and declines as the full width of the cell is attained behind the tip^{4,5}, as suggested in 1892 by Reinhardt⁶. Such a pattern of expansion is consistent with turgor-driven growth.

In this Review, I describe the roles of turgor and pressure in fungal growth (for previous reviews on the roles of turgor in mycelial fungi, see REFS 7,8). There are several key questions that are being researched in this field. For example, how is turgor regulated? How does it affect tip growth? Do intra-hyphal pressure gradients play a part in fungal growth (such as in the transport of new materials to the growing tip)? These areas of active research are revealing the mechanisms of hyphal growth in filamentous fungi and are relevant to applied research on pathogenicity and the control of fungal diseases.

The microbiological physics of pressure

The nature of turgor and its relevance for cellular expansion are founded on the ideal gas law (as molecules in the gaseous phase will be in equilibrium with the same molecular species in the liquid phase): $PV = nRT$ (in which P is pressure, V is cell volume, n is the number of moles of osmotically active solutes, R is the gas constant and T is temperature). Thus, pressure, volume and solute amounts are interrelated properties (temperature is usually assumed to remain constant) that are most

Department of Biology,
York University,
4700 Keele Street, Toronto,
Ontario M3J 1P3, Canada.
e-mail: planters@yorku.ca
doi:10.1038/nrmicro2591
Published online 6 June 2011

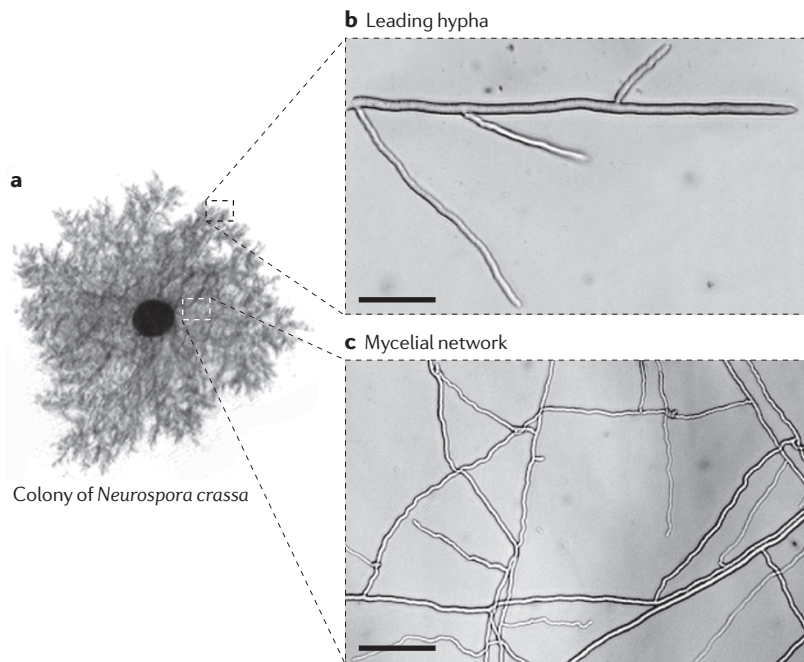


Figure 1 | Architecture of a fungal colony. **a** | From an initial inoculation with mycelium or spores, a fungal colony will radiate out, seeking to maximize its ability to obtain nutrients from the substrate. **b** | At the colony edge, leading hyphae are the first cells to invade new territory. They branch to maintain constant coverage of the substrate. **c** | Behind the colony edge, the hyphae branch and connect with other hyphae, thereby creating a network that optimizes the ability of the colony to extract nutrients from the substrate. Scale bars represent 100 μm .

easily explored in thermodynamics by invoking a closed system in which all but one of the properties are constant. However, in a growing cell, all these properties will change. For example, increases in volume decrease both solute concentration (given by n/V) and pressure, whereas increases in pressure result in increased volume and, therefore, decreased solute concentration. Complicating matters even more, these changes are affected by water flow into or out of the cell and by the extensibility of the cell wall (FIG. 2). To explore how these properties of the cell are interrelated during cellular expansion, the simplest approach is to consider them separately.

Pressure and volume. Changes in pressure affect volume, but the extent of the change depends on the resistance of the cell wall to deformation. This is described by the following equation: $dP/dV = \epsilon/V$ (in which ϵ is the volumetric modulus of cell wall elasticity). Changing either pressure (dP) or volume (dV) will cause the other to change; the size of the corresponding change depends on the initial volume and the elasticity of the wall. In walled cells, the modulus of elasticity increases with pressure^{9–14} (that is, at lower pressures, volume changes are relatively large). But as the pressure increases, the wall is no longer able to stretch, so the volume change is smaller. From an experimental perspective, optical measurements of hyphal diameter (from which relative hyphal volume can be calculated) offer an easy way to assess turgor changes (see BOX 1 for a summary of techniques that can be used

to measure turgor). However, the relationship between hyphal volume and turgor is not linear, owing to the pressure dependence of wall elasticity. Furthermore, any maintenance of turgor during cell expansion is affected by water flow into the cell.

Water flow. The dynamics of pressure and volume changes will depend on the rate of water flow into (or out of) the cell. The rate of change in pressure or volume depends on the hydraulic conductivity of the cell membrane and wall; fast water flow through a highly conductive membrane results in fast volume changes, whereas at a low hydraulic conductivity, the volume may not change at all. From the viewpoint of membrane biophysics, hydraulic conductivities are seldom thought to be limiting, because of the high permeation of water through membranes. But there must be instances in which water flow through membranes can be limiting, as indicated by the discovery of aquaporins¹⁵ and by their presence and expression in walled cells¹⁶. However, many fungal genomes lack aquaporin-encoding genes¹⁷. Furthermore, strains of *Saccharomyces cerevisiae* and *Candida albicans* in which the aquaporin genes have been mutated or deleted exhibit no discernible growth differences to wild type under normal conditions¹⁷. Only under the extreme condition of freeze–thaw are the aquaporins found to be beneficial to yeasts¹⁷. When rapid freezing occurs, extracellular water freezes first, while the intracellular water becomes super-cooled. Aquaporins enhance the rapid outflow of water from the cell and thereby inhibit the formation of lethal ice within the yeast cell. In the filamentous fungus *Neurospora crassa*, comparisons of hyperosmosis-induced hyphal volume shrinkage and of hypo-osmosis-induced hyphal tip lysis revealed no differences between wild type and an aquaporin-knockout strain (R.R.L., S. D. McClure and M.-S. Lim, unpublished observations). Thus, during tip expansion under normal conditions that are conducive to hyphal growth, water flows freely into the expansion zone without the assistance of aquaporins.

During normal growth, it is expected that water inflow will be maximal at the site of expansion and will taper off exponentially behind the growing tip (FIG. 2), because water inflow at the expanding tip attenuates the volume changes further behind the tip. Expansion at the tip may be regulated by wall extensibility.

Extensibility. Extensibility is a measure of how much the cell wall will expand as the pressure inside the cell is increased. In walled cells, extensibility varies with pressure: at low pressure, the wall is able to expand to a considerable extent, but at higher pressure it becomes stiff and inextensible — a prelude to structural failure and bursting. Although pressure is the driving force for tip extension, the rate of growth at the tip depends on the extensibility of the wall (and, crucially, the supply of cell wall¹⁸ and membrane material for the expanding tip). Expansion of the tip wall can be elastic or plastic. Elastic expansion is reversible (the stretched wall can revert to its original state, if turgor declines), whereas plastic expansion is irreversible (the wall shape has changed and

Aquaporins
Protein channels that transport water across cell membranes.

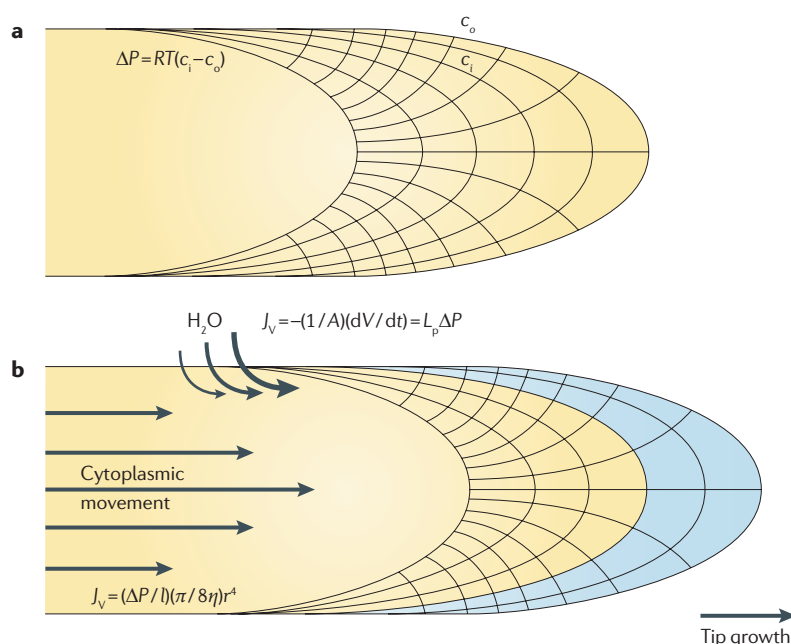


Figure 2 | Pressure, volume, water flow and hyphal growth. a | The hydrostatic force that expands the extensible tip is governed by the difference in osmotically active solutes inside and outside the cell, as given by the equation $\Delta P = RT(c_i - c_o)$, in which P is the pressure, R is the gas constant, T is the temperature, and c_i and c_o are the concentrations of osmotically active solutes inside and outside the cell, respectively. **b** | During expansion of the hyphal tip, water flow (J_v) into the cell (curved arrows) depends on the change in volume over time (dV/dt ; in which V is volume and t is time), and the area (A) through which the water can flow, as described by the formula $J_v = -(1/A)(dV/dt)$. The water flow can also be described according to the hydraulic conductivity, L_p , of the membrane and the pressure difference caused by the volume increase: $J_v = L_p \Delta P$. It is likely that water inflow will be greatest at the expanding tip, declining exponentially behind the tip. With water inflow, c_i will decline (and thus ΔP will decrease); so, steady-state growth requires the maintenance of turgor through the uptake of solutes and the biosynthesis of osmotically active solutes within the cell. Migration of the cytoplasm along with the growing tip can depend on the small intra-hyphal pressure difference that is created by tip expansion. The volume flow of the cytoplasm is described by the Poiseuille function, $J_v = (\Delta P/l)(\pi/8\eta)r^4$, in which $\Delta P/l$ is the intra-hyphal gradient (change in pressure divided by the length, l), η is the viscosity and r is the hyphal radius. Organellar movements (not shown) may also be mediated by the coordinated action of the cytoskeleton and molecular motors. See REF. 110 for a more rigorous analysis of the relationships between pressure, volume, water flow and cell growth.

Oomycete

A hyphal microorganism that is morphologically similar to fungi but belongs to the phylogenetically distinct stramenopiles group.

Characeae

A family of freshwater algae, within the kingdom Plantae. Members of this family form large macroscopic filaments.

Pectin

A carbohydrate polymer (a polysaccharide) that is part of the cell wall in walled cells of many organisms, such as algae and plants; pectin crosslinks can play a part in controlling wall strength and extensibility.

cannot revert to its original dimensions). Cell growth requires plastic deformation, which is achieved by the incorporation of new wall and membrane material at the expanding tip. The relationship between growth rate and wall softness (measured by the pressure at which the tip will burst) was explored in *Achlya bisexualis*, a hyphal oomycete¹⁹. These studies supported the classic relationship proposed by Lockhart²⁰: that growth rate depends on extensibility and a pressure that is above a threshold level (a minimum pressure below which cell expansion will not occur). That is, plastic deformation of the wall requires some minimum turgor, below which cell expansion will not occur. The interplay between turgor and extensibility has also been explored by researchers working on cell growth in algae and, to a lesser extent, higher plants. Experimentally, it is possible to clamp the pressure inside the cell at a specified value by impaling the cell with a pressure probe and modifying the

turgor (BOX 1). Increasing the turgor results in an initial increase in the elongation rate, followed by a return to the original rate despite the pressure being maintained at an elevated level; thus, changes in wall extensibility are used to maintain a constant growth rate.

But how does extensibility change? The best experimental evidence concerning changes in wall extensibility comes from research on the giant Characeae algae of the genera *Chara* and *Nitella*. Because of the large cell size of these organisms, the measurement and manipulation of turgor is simple²¹. Normal turgor in *Chara* spp. is about 600 kPa. When turgor is increased using the pressure probe, growth accelerates, but only transiently, before reverting to its original rate²². The extensibility is regulated by pectin and calcium pectate in the wall²³, as well as by the tension created by turgor²⁴. In the wall, Ca^{2+} crosslinks the pectin, stiffening the wall. When new pectin is supplied during cell expansion, it competes for the Ca^{2+} , causing the wall to loosen and allowing continued expansion. The ability of Ca^{2+} -linked pectin to release Ca^{2+} , thereby loosening the wall, requires a threshold pressure. Thus, pectin, Ca^{2+} and turgor coordinately maintain the growth rate through wall extensibility in Characeae algae²⁵. In the hyphal oomycetes *A. bisexualis*, *Achlya ambisexualis* and *Saprolegnia ferax*, the secretion of endoglucanase at high external osmolarities may have a similar role in regulating wall extensibility to maintain normal growth rates when turgor is low²⁶. In lily pollen tubes, growth rate and turgor are not correlated^{27,28}, and direct measurements of cell wall stiffness using a micro-indentation technique underscore the role of extensibility as the primary regulator of extension rates²⁹. Micro-indentation measurements use a fine glass fibre that touches the wall; the deflection of the fibre is used to determine the force that is required to indent the wall. This technique could also be used to explore the viscoelastic properties of the fungal wall, in concert with detailed explorations of the relationship between fungal wall composition and hyphal tip growth, to determine whether the regulation of extensibility plays a part in tip growth similarly to the situation in Characeae algae.

During hyphal growth in fungi and oomycetes, an elevated Ca^{2+} concentration at the hyphal tip regulates the growth of the tip. The mechanisms creating the Ca^{2+} gradient vary depending on the organism, but in all cases studied it involves a signal transduction pathway that detects changes in wall tension. In the hyphal oomycete *S. ferax*, Ca^{2+} is transported into the cell by stretch-activated Ca^{2+} channels at the tip³⁰ (this is also true for plant pollen tubes³¹). However, in the fungus *N. crassa*, Ca^{2+} influx is not preferentially located at the tip³², and stretch-activated Ca^{2+} channels do not control cell expansion, as their inhibition does not impair growth^{33,34}. Instead, the Ca^{2+} gradient is generated and maintained internally³⁵; inositol-1,4,5-trisphosphate (InsP_3) mediates the release of Ca^{2+} from internal stores at the tip, and this causes fusion of vesicles at the expanding tip. The InsP_3 seems to be produced by a stretch-activated phospholipase C^{36} . The Ca^{2+} is sequestered behind the tip by the endoplasmic reticulum (by the action of a Ca^{2+} -ATPase pump)^{37,38} and by tip-localized mitochondria³⁹. However,

Box 1 | Measuring cell pressure

There are a variety of techniques for measuring turgor in walled cells⁷⁶. The simplest of these is the addition of a hyperosmotic solution outside the cell. As water leaves the cell, the cell shrinks, and eventually the protoplast pulls away from the wall; this is known as plasmolysis. The process is observed through a microscope, and the percentage of plasmolysed cells is plotted versus the osmolarity of the osmotically active substance, to yield a measure of the threshold osmolarity that causes plasmolysis. The technique is fairly easy to use, but there are some caveats. The first is that the osmolyte must not be able to penetrate the cell, otherwise it will change the internal osmolarity. Moreover, scoring for plasmolysis is subjective, so careful researchers will use a 'blind' evaluation, scoring plates not knowing what treatment has been applied.

Another technique, osmometry, involves isolation of the internal cell sap (both cytoplasm and vacuolar contents) and measurement of its osmolarity relative to that of the external milieu. This is normally carried out using commercially available osmometers (which measure osmolarity as osmotically active moles per kg of solvent). At equilibrium — when the water potential is the same inside the cell as it is outside — the sap osmolarity is equal to the turgor. However, it is often difficult to ensure that the cell sap is not contaminated by the external medium (which can 'cling' to the cell walls), so this is not a popular technique.

A third method, the pressure probe, uses a fine glass 'needle' filled with an immiscible liquid; low-viscosity silicon oil is common. The needle is attached to a holder containing a pressure transducer. When the cell is impaled with the needle, turgor will force the oil interface into the needle. The amount of pressure that must be applied to 'push' the oil interface back to the tip is a measure of the cell turgor. The pressure probe measures the cell turgor directly but may cause damage to the cell as a consequence of impalement. When the pressure probe is used to measure turgor of filamentous fungi, it is helpful to monitor the cell under the microscope to ensure that there is no cell damage (see, for example, main text FIG. 3).

knockout mutants for some genes encoding Ca^{2+} transporters (*nca-1*, *nca-2*, *nca-3* and *cax*) exhibit a normal growth morphology, indicating that these genes are not essential for hyphal tip growth⁴⁰.

Thus, the tip-localized Ca^{2+} gradient is maintained during cell expansion by a variety of mechanisms. In all cases, stretching of the membrane and wall results in an elevated cellular Ca^{2+} concentration, which then mediates vesicle fusion to allow continued cell expansion and cell wall synthesis^{41,42}. The mechanism underlying Ca^{2+} -mediated vesicle fusion in fungi is not understood, but in animals it relies on several proteins, such as those belonging to the SNARE family of proteins⁴³. SNAREs have been identified in *N. crassa* and are localized to hyphal tips⁴⁴. Inactivation of the genes *syn1* and *syn2*, which encode two of these SNARE members, impair hyphal tip growth⁴⁵. A complete picture of how elevated Ca^{2+} could mediate this process remains to be obtained by further research.

Turgor regulation

With successful cell expansion, turgor will decline, but it must be maintained to provide a constant driving force during hyphal growth. In the natural environment, fungi may face wide extremes in extracellular osmolarity. These can result from changes in water availability as substrates dry out or the sudden osmotic shock of rain, which is a hypo-osmotic stress that could cause the internal hydrostatic pressure to rise to the point of causing cell wall rupture and cell lysis. To sustain growth (in the presence or absence of osmotic shock), fungi must

be able to adjust the concentrations of solutes internally to keep turgor at an acceptable level, or be able to modify cell wall extensibility.

Osmoresponses. Turgor provides a driving force for cell expansion in many organisms, and therefore it may need to be regulated. However, it should be noted that some hyphal organisms (for example, *A. bisexualis*) are not active turgor regulators⁴⁶ (see below). Turgor regulation can be explored directly by challenging the cells with an osmotic shock that causes a change in turgor. Normally, this is done using an impermeant substance (that is, an osmolyte that cannot enter the cell and thus cannot contribute to turgor recovery). When a permeant solute is added to the extracellular medium, the cell initially loses water (and so pressure declines), but uptake of the solute causes the intracellular osmolarity to increase, thus rebalancing the turgor. In the case of an impermeant solute, the cell must regulate turgor on its own, by taking up ions and synthesizing internal osmolytes such as glycerol. In *N. crassa*, turgor recovery takes about 10 min after treatment with permeant NaCl, compared with 60–90 min when the impermeant osmolyte sucrose is used⁴⁷. After sucrose treatment in *N. crassa*, turgor declines and a decrease in hyphal volume occurs concurrently with a transient depolarization of the membrane potential (all within about 2 min), followed by a rapid cessation of hyphal growth⁴⁸ (FIG. 3). The membrane potential then recovers and hyperpolarizes after 4 min. The sustained hyperpolarization to a potential that is significantly more negative than the initial potential occurs concomitantly with an increased net outward flux of H^+ , consistent with activation of the plasma membrane H^+ -ATPase⁴⁷. The hyperpolarized potential is presumably the driving force for the subsequent net uptake of ions (K^+ and Cl^-) followed by turgor recovery⁴⁷. The synthesis of glycerol occurs after about 20 min⁴⁹.

The osmotic mitogen-activated protein kinase cascade.

The two mechanisms for turgor regulation in fungi (ion accumulation and osmolyte synthesis) are controlled by an osmotic mitogen-activated protein kinase (MAPK) pathway⁴⁷. In general, MAPK pathways comprise a cascade of kinases that are sequentially activated by phosphorylation to amplify a transduction signal. The kinases are usually known as MAPK, MAPK kinase (MAPKK) and MAPKK kinase (MAPKKK) to indicate their position in the sequential cascade. The MAPK pathway is activated by external stimuli acting on an upstream receptor, and the best studied consequence of this activation is the expression of pathway-specific genes. In fungi, MAPK cascades have roles in different processes, such as the responses to external pheromones in mating and the responses to various stresses.

The osmotic MAPK cascade was initially characterized in *S. cerevisiae*⁵⁰ and appears to be ubiquitous in fungi⁵¹, although variation occurs in the components of the upstream cascade^{52–56}. In a number of fungal species, the osmotic MAPK pathway proteins were initially identified because mutations in the encoding genes resulted in osmosensitive mutants. In *N. crassa*, the osmotic MAPK

SNARE

(Soluble NSF (N-ethylmaleimide-sensitive factor) attachment protein (SNAP) receptor). A family of receptors that are involved in membrane fusion in animal cells.

pathway is composed of OS-1 (the sensor), OS-4 (the MAPKKK), OS-5 (the MAPKK) and OS-2 (the MAPK). Inactivation of any of the corresponding genes leads to osmosensitive mutants, and mutants for either *os-1* or *os-2* have lower turgor than wild type and exhibit incomplete turgor recovery in response to hyperosmotic shock⁴⁷. That these mutants have lower turgor than wild type even under normal conditions suggests that the MAPK cascade operates even under non-stress conditions.

The cascade controls ion transport (that is, H^+ pump activity and K^+ uptake) as part of the osmotic response of the cell and does so even in the absence of *de novo* protein synthesis⁴⁷ (FIG. 4). The electrical response to hyperosmotic treatment of *N. crassa* — that is, hyperpolarization — and subsequent ion uptake are not observed in mutants for the osmotic MAPK cascade. The common description of the cascade as the high osmotic glycerol (HOG) pathway, referring to glycerol synthesis after treatment with high levels of salt in *S. cerevisiae*, is unfortunate. Certainly, activation of the osmotic MAPK cascade by osmotic shock causes the synthesis of osmolytes (glycerol in *N. crassa*⁵⁷), but turgor is regulated even in the absence of glycerol synthesis by the activation of ion uptake, as demonstrated for the osmosensitive *cut* mutant of *N. crassa*⁴⁹ (*cut-1* encodes a protein of unknown function that belongs to the haloacid dehalogenase family⁵⁸ and is unrelated to the osmotic MAPK cascade). The central role of the osmotic MAPK cascade in the regulation of ion transport is supported by the observation that ectopic activation of the cascade with the phenylpyrrole fludioxonil (which activates OS-1, the sensor^{59,60}) causes not only glycerol synthesis⁶¹, but also activation of the H^+ -ATPase and ion uptake⁶².

Besides the osmotic MAPK cascade, other proteins have a role in turgor regulation, probably as part of separate transduction pathways. *N. crassa* knockout mutants of the genes *mid-1* and *ptk-2* (REF. 63), which encode a putative stretch-activated cation channel and a kinase regulator of H^+ -ATPase activity, respectively, exhibit lower turgor and lower H^+ -ATPase activity than wild type under standard growing conditions^{64,65} (FIG. 4).

Non-turgid growth. The central role of turgor in hyphal growth is not universal. Some hyphal organisms, such as the oomycetes *S. ferax* and *A. bisexualis*, do not regulate turgor^{19,46,66} and therefore may not rely solely on turgor to drive growth. In fact, tip growth in *S. ferax* can occur in the absence of measurable turgor⁶⁷. As the life cycles of *S. ferax* and *A. bisexualis* have motile-cell stages during which contractile vacuoles are used for osmoregulation, the osmoregulatory mechanisms of these oomycetes may be different from those of fungi. In *A. bisexualis*, invasively and non-invasively growing hyphae have similar turgor and burst pressures⁶⁸ but different distributions of actin, which suggests that the cytoskeleton plays a part in invasive growth in this organism.

Furthermore, even in a turgor-regulating organism such as *N. crassa*, only a few mutations are required to create an amoeboid cell (a mutant called *slime*^{69,70}), with pseudopodial-like extensions that are presumably driven by the cytoskeleton. The *slime* mutant lacks a cell

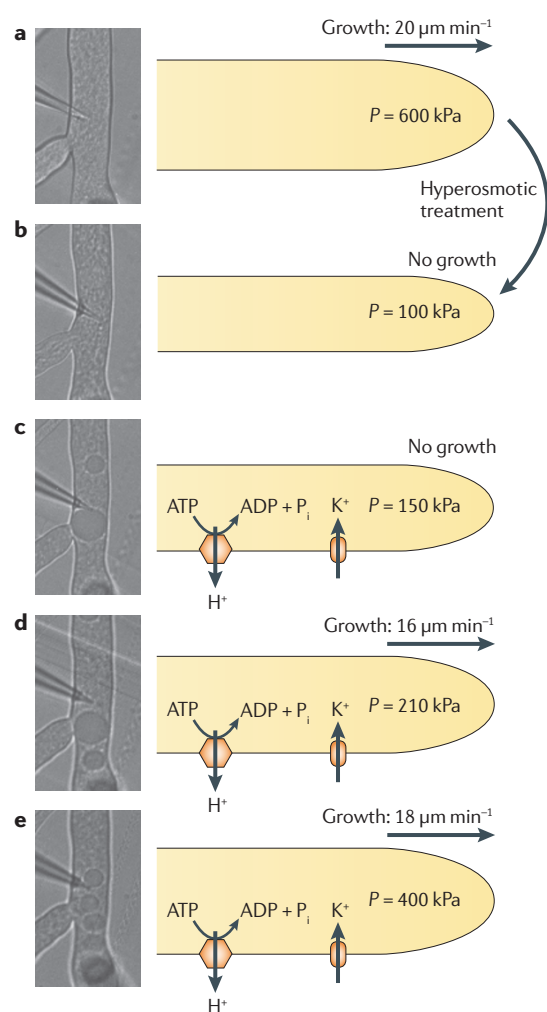


Figure 3 | Pressure, volume and its regulation. The data shown here were obtained from time courses in which hyperosmotic treatment was used to decrease the turgor in *Neurospora crassa* and the process of turgor recovery was monitored over 70 min using a pressure probe⁹. The photographs show an example of a hypha impaled with a pressure probe and the changes in hyphal diameter that were caused by hyperosmotic treatment. The spherical structures inside the impaled hyphae are droplets of silicon oil, which move away from the impalement site as a consequence of cytoplasmic flow. **a** | Before treatment, turgor (or pressure, P) is high (600 kPa) and the hypha is growing at $20 \mu\text{m min}^{-1}$. **b** | Turgor and relative hyphal volume decrease immediately after hyperosmotic treatment, soon followed by growth arrest. **c–e** | The H^+ -ATPase is activated, followed by ion (K^+) uptake into the hypha. As a result, the electrical potential becomes more negative and the internal osmolarity increases. Turgor recovery is observed, followed by the resumption of hyphal growth. Glycerol synthesis also occurs (not shown), but lags behind turgor recovery.

wall⁷¹ and can be lysed with osmotic treatment⁷², so turgor in these cells, if there is any, should be low. Clearly, growth and substrate penetration may not depend solely on turgor, and alternative tip growth patterns and processes can exist⁷³.

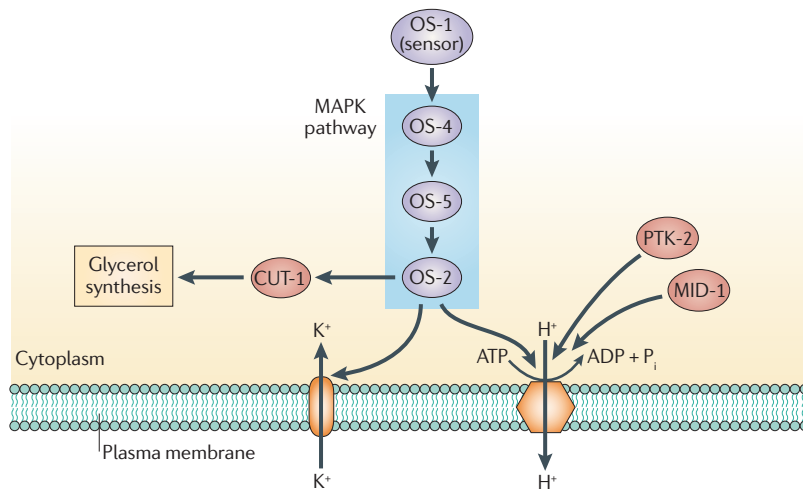


Figure 4 | Turgor recovery mechanisms in *Neurospora crassa*. A model of the roles of the osmotic mitogen-activated protein kinase (MAPK) cascade, MID-1 (a stretch-activated cation channel), PTK-2 (a kinase regulator of the H⁺-ATPase) and CUT-1 (a protein of unknown function from the haloacid dehalogenase family; a CUT-1-defective mutant is osmosensitive and does not synthesize glycerol). The MAPK cascade, which is activated by the action of the osmosensor OS-1, is a central control point for turgor recovery and stimulates ion uptake (which, alone, is sufficient for turgor recovery) and glycerol synthesis. PTK-2 and MID-1 function in separate pathways that contribute to turgor under normal growing conditions.

Mass flow of cytoplasm

Microscopic observation of mycelial colonies often reveals the active movement of cytoplasm through the hyphal network, usually towards growing hyphae at the colony edge. However, the nature of the apparent cytoplasmic flow cannot be discerned by microscopic observation, because movements mediated by the cytoskeleton and associated molecular motors might give the optical impression of mass movement of cytoplasm. Mass (or cytoplasmic) flow is defined as the movement of the internal volume within the hypha, similar to water flow in a pipe. Direct evidence of mass flow comes from experiments in which silicon oil (which would not provide any binding sites for the molecular motors) was injected into hyphae; the oil globules were found to move similarly to larger organelles, such as vacuoles, indicating that mass flow was occurring⁷⁴.

How is this mass movement of cytoplasm produced? Early research in *N. crassa* reported important apicobasal differences in osmotic pressures, as measured using an incipient plasmolysis technique (BOX 1), that would support a pressure-driven migration of cytoplasm towards the growing tips at the colony edge: apical pressures of 1.26 MPa versus basal pressures of 1.77 MPa⁷⁵. However, such pressure differentials have not been observed using the more direct pressure probe methods, which report much lower turgor (around 500 kPa) and no measurable pressure gradient. Reasons for the discrepancy are not clear, although scoring for incipient plasmolysis is subjective and it is difficult to correct for volume changes of the hyphae and differing cell wall extensibility of apical and basal cells^{75,76}. However, small pressure gradients (0.05–10 kPa cm⁻¹) would, in theory, be enough to explain the mass flow of cytoplasm if the flow is laminar

(see BOX 2). Such small pressure differences cannot be measured directly with the pressure probe.

An alternative approach that can be used to demonstrate the role of a pressure differential in cytoplasmic flow is to modify the intra-hyphal pressure gradient by local hyperosmotic treatment to pull water out of the hyphae at a localized region, thereby generating a low-turgor region in the hyphal network. When it is placed in front of the colony edge, the hyperosmotic treatment causes a pressure drop at the colony edge, and mass flow rates towards the colony edge increase, whereas when the treatment is placed behind the observation point, so that the pressure drop occurs behind the colony edge, mass flow rates towards the colony edge decrease⁷⁴ (FIG. 5). Put simply, experimental modification of the intra-hyphal pressure gradient modifies flow in a manner that is consistent with mass flow. Therefore, these studies indicate that hyphal networks are micro-hydraulic networks. Regulation of flow involves the closure of septa, which become plugged by Woronin bodies in filamentous ascomycetes. Modifications of the protein that tethers the Woronin body at the septum affect the ability of the hypha to plug the septa after wounding⁷⁷. It has been proposed that, in addition to a role protecting hyphae from wounding, septa may regulate the rates of cytoplasmic mass flow that in turn regulate the rates of hyphal growth at the colony edge⁷⁸. The concept of growth-induced mass flow in mycelial networks⁷⁹ is a promising perspective from which to build future experimental explorations.

Molecular motors and mass flow: the great divide? The respective roles of mass flow and of active transport via molecular motors in the delivery of building materials to the hyphal tip are difficult to assess because the different biophysical and biochemical regimes that are involved in these two processes require the use of different experimental approaches. Mass flow operates over long distances; molecular motors do not. How these two processes relate to hyphal growth is also quite different.

Models of hyphal growth that focus exclusively on either molecular motors or turgor have been devised. Sugden *et al.*⁸⁰ presented a physical model of vesicles being carried by kinesin molecular motors along microtubule tracks that extend into the tip. Such a model of vesicles hopping along multiple one-dimensional lattices could supply the necessary vesicles and could be self-regulating (should tip growth be halted for whatever reason, a 'traffic jam' would ensue). By contrast, Boudaoud⁸¹ described a model that focused on how turgor and tension interact to 'shape' the growing cell. Thus, both molecular motors and turgor are essential, but integrating the two into a cohesive growth mechanism is challenging.

With the availability of mutants and fast methods for creating gene knockouts in *N. crassa*⁶³, and with tools for live-cell fluorescence imaging^{82–84}, it has become possible to explore hyphal growth and the role of the cytoskeleton in greater detail. For instance, an *N. crassa* actin mutant exhibits aberrant morphology (apical branching), but the general tip-growing form of hyphae is retained⁸⁵. The knockout is available only as a heterokaryon, so the gene is probably essential. In *N. crassa*, mutations in

Woronin bodies

Proteinaceous granules that are located at septal pores and can seal the pores in response to cell damage.

Kinesin

A molecular motor that normally moves towards the plus end of microtubules, energized by ATP hydrolysis.

Heterokaryon

A hypha that contains multiple nuclei which are genetically distinct.

Box 2 | Microfluidic mechanics

Mass flow of a liquid through a tube can be either laminar (smoothly transitioning from no flow at the edges to maximal flow at the centre) or turbulent (chaotic, with vortices). The type of flow can be predicted by calculation of the Reynolds number (Re), which is defined by the equation $Re = (\rho vd) / \eta$ (in which ρ is the density, v is the velocity of flow, d is the tube diameter and η is the viscosity). For flow through a tube, the flow will be laminar when $Re < 10^3$, and the pressure gradient in X direction (dP/dx) that is required for flow can be calculated from the following equation (known as the Poiseuille equation): $dP/dx = 4v\eta/r^2$ (in which r is the radius of the tube). In hyphae of *Neurospora crassa*, Re is about 10^{-4} , so turbulent flow does not occur. In this case, the pressure gradient required for flow (calculated from the Poiseuille equation) is in the range of 0.0005 – 0.1 bar cm^{-1} (that is, 0.05 – 10 kPa cm^{-1})⁷⁴, which is similar to the values observed for microfluidic devices¹⁰⁵. These pressure gradients are very low compared with the normal turgor of the hyphae (430 – 570 kPa^{46,47,49,65}) and, owing to the experimental variability of turgor measurements, cannot be demonstrated with the pressure probe technique.

Because the flow of the hyphal cytoplasm is laminar, there is no turbulence-mediated mixing. Instead, mixing must occur through diffusion¹⁰⁵. To give a quantitative explanation of the intertwining of mass flow and diffusive flow, another dimensionless number — the Péclet number (Pe), or the ratio of velocity-driven mixing to diffusive mixing — provides a basic test of the importance of the two types of mixing^{106–109}. This number is defined by the equation $Pe = vd/D$ (in which D is the diffusion coefficient). For a protein in a fungal hypha, the diffusion coefficient is about $7 \times 10^{-11} m^2 s^{-1}$, the flow velocity is about $5 \times 10^{-6} m s^{-1}$ and the tube diameter is about $15 \times 10^{-6} m$ (REF. 74). So, Pe is about 1 and, therefore, both flow and diffusion contribute equally. For metabolites smaller than a protein, diffusion is more important, as they diffuse more quickly. Diffusion of organelles would have a higher Pe (because of their smaller diffusion coefficient); that is, flow of organelles is more important than diffusive mixing in a fungal hypha. In any case, organelles are often transported with the assistance of molecular motors.

the genes encoding the microtubule-related motors dynactin and dynein affect the distribution of nuclei (which are basally located but still migrate with the growing tip) and affect the 'straightness' of hyphal growth (the hyphae zig-zag as they grow)^{86–88}. In these mutants, the transport of nuclei along with the growing tip still occurs, even in the absence of a functional microtubule system, presumably owing to mass flow⁸⁹. The movement of nuclei through hyphal fusions between conidial germlings (newly germinated conidia) is still observed in microtubule mutants of *N. crassa*, but whether this can be ascribed to mass flow is unclear⁹⁰.

What should be emphasized is the intrinsic mechanical difference between mass flow and transport that is mediated by molecular motors. In the case of motor-mediated transport, only the bound cargo is moved: owing to the low-Reynolds-number environment, the cargo will not 'drag' the surrounding cytoplasm with it (BOX 2). Evidence in support of this assertion comes from experiments in which inhibition of cytoplasmic particle streaming has no effect on the movement of cytoplasm-located fluorescent dyes through coupled cells⁹¹. In the case of mass flow, there is no discrimination: anything that is not anchored will be transported.

Macroscale polarity. There is no doubt that there is an intrinsic polar distribution of cytological elements and ion transporters at the growing hyphal apex: there is a microscale polarity of tip-enriched actin^{92,93}, tip-localized mitochondria³⁹ and the Spitzenkörper⁹⁴. But macroscale polarity may also occur. Unlike molecular motors, mass flow operates over long distances (centimetres). In other words, small osmotic imbalances might create slight intra-hyphal pressure gradients, effecting mass movement over long distances. This raises the question of whether mass flow has a role in creating a macropolarity in the hyphal colony (FIG. 6). One can envision the leading hypha exploring new territory, while behind, in the

mycelial network, maximal uptake of nutrients occurs, leading to increased water inflow, increased pressure and a driving force that 'pushes' cytoplasm towards the colony edge. If this is so, then one would expect to see evidence of long-distance polarity in other cell processes, in particular in ion transport, which generates osmotic gradients that cause changes in turgor and could also create electrical currents.

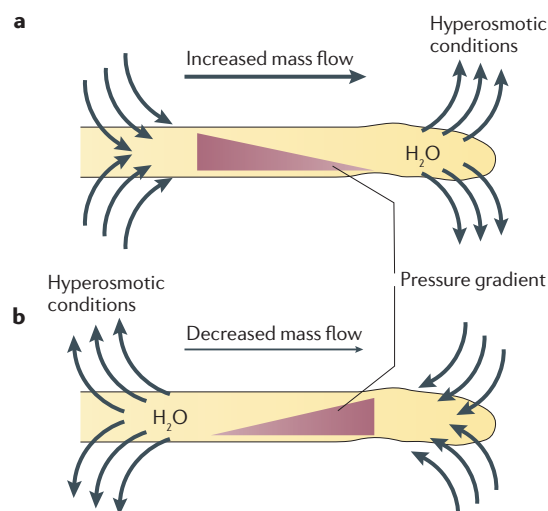


Figure 5 | Mass flow of the hyphal cytoplasm. Evidence for mass flow of the cytoplasm being driven by intra-hyphal pressure (measured 1–1.5 cm behind the colony edge) comes from experimental manipulation of intra-hyphal gradients by localized hyperosmotic treatment.

a | Hyperosmotic conditions at the hyphal tip cause efflux of water (curved arrows) from the tip and acceleration of the mass flow of cytoplasm towards the tip. **b** | Hyperosmotic treatment behind the tip causes water efflux behind the tip and reduction of the normal mass flow of cytoplasm towards the tip. Thus, the hyphal network acts as a network of microhydraulic tubes.

Spitzenkörper

An organized assembly of secretory vesicles located at the tip of the growing hypha.

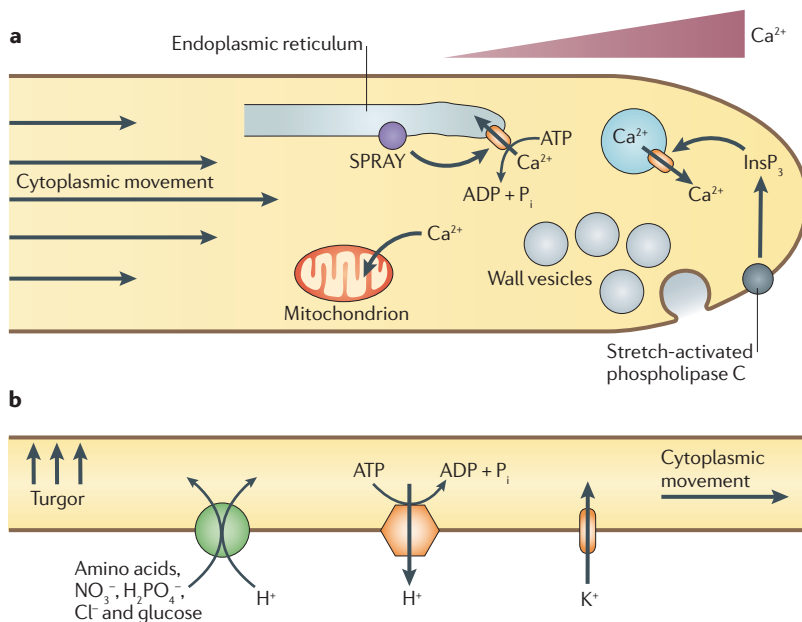


Figure 6 | Model of the polar distribution of transport in a mycelial colony.
a | At the growing tip, the major transport system may function to maintain the tip-high Ca^{2+} gradient. In *Neurospora crassa*, inositol-1,4,5-trisphosphate (InsP_3), which is probably produced by a stretch-activated phospholipase C, mediates the release of Ca^{2+} from internal stores at the tip, and then Ca^{2+} causes fusion of vesicles at the expanding tip. The Ca^{2+} is sequestered behind the tip by the endoplasmic reticulum (by the action of a Ca^{2+} -ATPase pump) and the mitochondria. SPRAY is a putative regulator of Ca^{2+} sequestration¹¹¹. **b** | Behind the tip (within the mycelial network), ion and nutrient uptake are driven by the activity of the plasma membrane H^+ -ATPase. Some of the known (and assumed, such as Cl^-) H^+ transport partners are shown, including amino acids¹¹², nitrate (NO_3^-)^{113,114}, phosphate (H_2PO_4^-)¹¹⁵ and glucose¹¹⁶. The intra-hyphal pressure gradient causes cytoplasmic movement into the tip.

It is possible to measure the electrical currents that surround organisms by using a vibrating probe. Indeed, electrical currents surrounding hyphae have been reported for several fungi; the electrical current is inwards at the tip and outwards basally⁹⁵. In *N. crassa*, the electrical current is carried principally by H^+ (REFS 96,97). Indirect support for the role of polar currents in normal growth comes from experiments documenting the effect of extracellular voltage fields on the direction of growth^{98,99}. However, it is unlikely that there is a single electrogenic transporter responsible for the polar currents. Net currents will depend on the ion species that are available for uptake; these species can include ammonium (NH_4^+) and nitrate (NO_3^-), which will affect other

fluxes owing to their impact on pH regulation, as will the transport of small metabolites¹⁰⁰. With respect to fluxes of specific ions (in this case, Ca^{2+} and H^+) measured with ion-selective probes, some hyphal organisms (such as the oomycete *S. ferax*) exhibit tip-localized ion currents, but others (such as the ascomycete *N. crassa*) do not³²; this difference is in part due to the asymmetrical localization of ion channels in *S. ferax* but not in *N. crassa*³³. The distally located plasma membrane H^+ -ATPase¹⁰¹ may play a key part in both generating the polar current patterns and creating the trans-hyphal pressure gradient, and is the likely cause of a depolarized potential that is commonly reported at the growing edge of the colony^{35,102,103}.

There is no universal model of microscale and macroscale polarity in hyphae, but the evidence does support the possibility of variations on a central theme of polarity. It should not be surprising that multiple disparate mechanisms — pressure and molecular motors — contribute to the growth of fungi. The use of proteins labelled with fluorescent tags may assist greatly in revealing the extent of microscale and macroscale polarity *in vivo*, extending previous cytochemical studies¹⁰⁴.

Concluding remarks

How does a hypha grow? Currently, we know only parts of the answer. Localized incorporation of vesicles and wall material creates the characteristic tip-growing form of the hypha. Cytoskeletal structures (actin, microtubules and associated molecular motors) organize the polar architecture of the growing tip. But microbiological physics is also important: pressure is a driving force for cellular expansion and growth. We are beginning to understand elements of the regulation of pressure and growth, revealed through the combination of mutant analysis and physical measurements of turgor. Regulation of wall extensibility is a key element that controls the rates of cell expansion, but much more needs to be elucidated concerning this form of regulation. Direct experimental measurements of the extensibility of hyphae would be the first step in a quantitative description that would assist the development of growth models. The nature of fungal microfluidics has been established, but we do not know how it integrates with other aspects of cell growth, and new methods of inquiry will be required to establish this. Radioisotope and fluorescent mapping of cytoplasmic flow are two techniques that may reveal how patterns of flow contribute to colony growth.

- Trail, F. Fungal cannons: explosive spore discharge in the Ascomycota. *FEMS Microbiol. Lett.* **276**, 12–18 (2007).
 - Yafetto, L. et al. The fastest flights in nature: high-speed spore discharge mechanisms among fungi. *PLoS ONE* **3**, e3237 (2008).
 - Bastmeyer, T., Deising, H. B. & Bechinger, C. Force exertion in fungal infection. *Annu. Rev. Biophys. Biomol. Struct.* **31**, 321–341 (2002).
 - Bartnicki-Garcia, S., Bracker, C. E., Gierz, G., Lopez-Franco, R. & Lu, H. Mapping the growth of fungal hyphae: orthogonal cell wall expansion during tip growth and the role of turgor. *Biophys. J.* **79**, 2382–2390 (2000).
- A careful analysis of the nature of tip expansion in growing hyphae by microscopic observation of both**

- external and internal markers. The results strongly support the idea that expansion is orthogonal. Biophysical considerations lead to the conclusion that turgor drives cell expansion.**
- Shaw, S. L., Dumais, J. & Long, S. R. Cell surface expansion in polarly growing root hairs of *Medicago truncatula*. *Plant Physiol.* **124**, 959–970 (2000).
 - Reinhardt, M. O. Das Wachstum der pilzhypen. *Jahrbücher für Wissenschaftliche Botanik* **23**, 479–566 (1892).
 - Money, N. P. in *The Mycota. I. Growth, Differentiation and Sexuality* (eds Wessels, J. G. H. & Meinhardt, F.) 67–88 (Springer, 1994).
 - Money, N. P. in *The Mycota. VIII. Biology of the Fungal Cell 2nd edn* (eds Howard, R. J. & Gow, N. A. R.) 237–249 (Springer, 2007).

- Lew, R. R. & Nasserifar, S. Transient responses during hyperosmotic shock in the filamentous fungus *Neurospora crassa*. *Microbiology* **155**, 903–911 (2009).
- Steudle, E., Zimmerman, U. & Lüttge, U. Effect of turgor pressure and cell size on the wall elasticity of plant cells. *Plant Physiol.* **59**, 285–289 (1977).
- Steudle, E., Zimmerman, U. & Zillikens, J. Effect of cell turgor on hydraulic conductivity and elastic modulus of *Elodea* leaf cells. *Planta* **154**, 371–380 (1982).
- Cosgrove, D. J. In defence of the cell volumetric elastic modulus. *Plant Cell Environ.* **11**, 67–69 (1988).
- Franks, P. J., Buckley, T. N., Shope, J. C. & Mott, K. A. Guard cell volume and pressure measured concurrently by confocal microscopy and the cell pressure probe. *Plant Physiol.* **125**, 1577–1584 (2001).

14. Ortega, J. K. E., Keanini, R. G. & Manica, K. J. Pressure probe technique to study transpiration in *Phycomyces sporangiotheca*. *Plant Physiol.* **87**, 11–14 (1988).
15. Agre, P. Aquaporin water channels. *Nobelprize.org* [online], http://nobelprize.org/nobel_prizes/chemistry/laureates/2003/agre-lecture.html (2003).
16. Tyerman, S. D., Bohnert, H. J., Maurel, C., Steudle, E. & Smith, J. A. C. Plant aquaporins: their molecular biology, biophysics and significance for water relations. *J. Exp. Bot.* **50**, 1055–1071 (1999).
17. Tanghe, A., Van Dijk, P. & Thevelein, J. M. Why do microorganisms have aquaporins? *Trends Microbiol.* **14**, 78–85 (2005).
18. Bartnicki-Garcia, S. Chitosomes: past, present and future. *FEMS Yeast Res.* **6**, 957–965 (2006).
19. Money, N. P. & Harold, F. M. Extension growth of the water mold *Achlya*: interplay of turgor and wall strength. *Proc. Natl Acad. Sci. USA* **89**, 4245–4249 (1992).
20. Lockhart, J. A. An analysis of irreversible plant cell elongation. *J. Theor. Biol.* **8**, 264–275 (1965).
21. Green, P. B. Growth physics in *Nitella*: a method for continuous *in vivo* analysis of extensibility based on a micro-manometer technique for turgor pressure. *Plant Physiol.* **43**, 1169–1184 (1968).
22. Zhu, G. L. & Boyer, J. S. Enlargement in *Chara* studies by a turgor clamp. Growth rate is not determined by turgor. *Plant Physiol.* **100**, 2071–2080 (1992).
23. Proseus, T. E. & Boyer, J. S. Calcium pectate chemistry controls growth rate of *Chara corallina*. *J. Exp. Bot.* **57**, 3989–4002 (2006).
24. Proseus, T. E. & Boyer, J. S. Tension required for pectate chemistry to control growth in *Chara corallina*. *J. Exp. Bot.* **58**, 4283–4292 (2007).
25. Proseus, T. E. & Boyer, J. S. Calcium pectate chemistry causes growth to be stored in *Chara corallina*: a test of the pectate cycle. *Plant Cell Environ.* **31**, 1147–1155 (2008).
26. Money, N. P. & Hill, T. W. Correlation between endoglucanase secretion and cell wall strength in oomycete hyphae: implications for growth and morphogenesis. *Mycologia* **89**, 777–785 (1997).
27. Benkert, R., Obermeyer, G. & Bentrup, F.-W. The turgor pressure of growing lily pollen tubes. *Protoplasma* **198**, 1–8 (1997).
28. Winship, L. J., Obermeyer, G., Geitmann, A. & Hepler, P. K. Under pressure, cell walls set the pace. *Trends Plant Sci.* **15**, 363–369 (2010).
29. Zerkow, R., Kroeger, J. & Geitmann, A. Polar growth in pollen tubes is associated with spatially confined dynamic changes in cell mechanical properties. *Dev. Biol.* **334**, 437–446 (2009).
- A detailed exploration of the relationship between the mechanical properties of the cell wall and tip growth. A micro-indentation technique is used to quantify wall stiffness, thereby integrating cytomechanics with cell expansion.**
30. Levina, N. N., Lew, R. R. & Heath, I. B. Cytoskeletal regulation of ion channel distribution in the tip-growing organism *Saprolegnia ferax*. *J. Cell Sci.* **107**, 127–134 (1994).
31. Dutta R. & Robinson, K. R. Identification and characterization of stretch-activated ion channels in pollen protoplasts. *Plant Physiol.* **135**, 1398–1406 (2004).
32. Lew, R. R. Comparative analysis of Ca²⁺ and H⁺ flux magnitude and location along growing hyphae of *Saprolegnia ferax* and *Neurospora crassa*. *Eur. J. Cell Biol.* **78**, 892–902 (1999).
33. Lew, R. R. Mapping fungal ion channel distributions. *Fungal Genet. Biol.* **24**, 69–76 (1998).
34. Levina, N. N., Lew, R. R., Hyde, G. J. & Heath, I. B. The roles of calcium ions and plasma membrane ion channels in hyphal tip growth of *Neurospora crassa*. *J. Cell Sci.* **108**, 3405–3417 (1995).
35. Silverman-Gavrila L. B. & Lew, R. R. Calcium and tip growth in *Neurospora crassa*. *Protoplasma* **213**, 203–217 (2000).
36. Silverman-Gavrila L. B. & Lew, R. R. Calcium gradient dependence of *Neurospora crassa* hyphal growth. *Microbiology* **149**, 2475–2485 (2003).
- A description of the important role that intracellular Ca²⁺ gradients have in fungal tip growth.**
37. Silverman-Gavrila L. B. & Lew, R. R. Regulation of the tip-high [Ca²⁺] gradient in growing hyphae of the fungus *Neurospora crassa*. *Eur. J. Cell Biol.* **80**, 379–390 (2001).
38. Silverman-Gavrila L. B. & Lew, R. R. An IP₃-activated Ca²⁺ channel regulates fungal tip growth. *J. Cell Sci.* **115**, 5013–5025 (2002).
39. Levina, N. N. & Lew, R. R. The role of tip-localized mitochondria in hyphal growth. *Fungal Genet. Biol.* **43**, 65–74 (2006).
40. Bowman, B. J., Abreu, S., Margoles-Clark, E., Draskovic, M. & Bowman, E. J. Role of four calcium transport proteins, encoded by *nca-1*, *nca-2*, *nca-3*, and *cax*, in maintaining intracellular calcium levels in *Neurospora crassa*. *Eukaryot. Cell* **10**, 654–661 (2011).
41. Verdin, J., Bartnicki-Garcia, S. & Riquelme, M. Functional stratification of the Spitzenkörper of *Neurospora crassa*. *Mol. Microbiol.* **74**, 1044–1053 (2009).
42. Riquelme, M. *et al.* Spitzenkörper localization and intracellular traffic of green fluorescent protein-labeled CHS-3 and CHS-6 chitin synthases in living hyphae of *Neurospora crassa*. *Eukaryot. Cell* **6**, 1853–1864 (2007).
43. Hilfiker, S., Greengard, P. & Augustine, G. J. Coupling calcium to SNARE-mediated synaptic vesicle fusion. *Nature Neurosci.* **2**, 104–106 (1999).
44. Gupta, G. D. & Heath, I. B. A tip-high gradient of a putative plasma membrane SNARE approximates the exocytotic gradient in hyphal apices of the fungus *Neurospora crassa*. *Fungal Genet. Biol.* **29**, 187–199 (2000).
45. Gupta, G. D., Free, S. J., Levina, N. N., Keränen, S. & Heath, I. B. Two divergent plasma membrane syntaxin-like SNAREs, *nsyn* and *nsyn2*, contribute to hyphal tip growth and other developmental processes in *Neurospora crassa*. *Fungal Genet. Biol.* **40**, 271–286 (2003).
46. Lew, R. R., Levina, N. N., Walker, S. K. & Garrill, A. Turgor regulation in hyphal organisms. *Fungal Genet. Biol.* **41**, 1007–1015 (2004).
47. Lew, R. R., Levina, N. N., Shabala, L., Anderca, M. I. & Shabala, S. N. Role of a mitogen-activated protein kinase cascade in ion flux-mediated turgor regulation in fungi. *Eukaryot. Cell* **5**, 480–487 (2006).
- A suite of techniques are used to demonstrate the role of ion transport in turgor regulation, which is mediated by a MAPK cascade, in a fungus.**
48. Lew, R. R. & Nasserifar, S. Transient responses during hyperosmotic shock in the filamentous fungus *Neurospora crassa*. *Microbiology* **155**, 903–911 (2009).
49. Lew, R. R. & Levina, N. N. Turgor regulation in the osmosensitive *cut* mutant of *Neurospora crassa*. *Microbiology* **153**, 1530–1537 (2007).
50. Brewster, J. L., de Valoir, T., Dwyer, N. D., Winter, E. & Gustin, M. C. An osmosensing signal transduction pathway in yeast. *Science* **259**, 1760–1763 (1993).
51. Kranz, M., Becit, E. & Hoffmann, S. Comparative genomics of the HOG-signaling system in fungi. *Curr. Genet.* **49**, 137–151 (2006).
52. Maeda, T., Takekawa, M. & Saito, H. Activation of yeast PBS2 MAPKK by MAPKKs or by binding of an SH3-containing osmosensor. *Science* **269**, 554–558 (1995).
53. Posas, F. & Saito, H. Osmotic activation of the HOG MAPK pathway via Ste11p MAPKKK: scaffold role of Pbs2p MAPKK. *Science* **276**, 1702–1705 (1997).
54. Alex, L. A., Borkovich, K. A. & Simon, M. I. Hyphal development in *Neurospora crassa*: involvement of a two-component histidine kinase. *Proc. Natl Acad. Sci. USA* **93**, 3416–3421 (1996).
55. Schumacher, M. M., Enderlin, C. S. & Selitrennikoff, C. P. The osmotic-1 locus of *Neurospora crassa* encodes a putative histidine kinase similar to osmosensors of bacteria and yeast. *Curr. Microbiol.* **34**, 340–347 (1997).
56. Jones, C. A., Greer-Philips, S. E. & Borkovich, K. A. The response regulator RRG-1 functions upstream of a mitogen-activated protein kinase pathway impacting asexual development, female fertility, osmotic stress, and fungicide resistance in *Neurospora crassa*. *Mol. Biol. Cell* **18**, 2123–2136 (2007).
57. Ellis, S. W., Grindle, M. & Lewis, D. H. Effect of osmotic stress on yield and polyol content of dicarboximide-sensitive and -resistant strains of *Neurospora crassa*. *Mycol. Res.* **95**, 457–464 (1991).
58. Youssar, L. & Avalos, J. Genetic basis of the *ovc* phenotype of *Neurospora*: identification and analysis of a 77kb deletion. *Curr. Genet.* **51**, 19–30 (2007).
59. Ochial, N. *et al.* Characterization of mutations in the two-component histidine kinase gene that confer fludioxonil resistance and osmotic sensitivity in the *os-1* mutants of *Neurospora crassa*. *Pest Manag. Sci.* **57**, 437–442 (2001).
60. Motoyama, T. *et al.* An Os-1 family histidine kinase from a filamentous fungus confers fungicide-sensitivity to yeast. *Curr. Genet.* **47**, 298–306 (2005).
61. Fujimura, M. *et al.* Sensitivity to phenylpyrrole fungicides and abnormal glycerol accumulation in *Os* and *Cut* mutant strains of *Neurospora crassa*. *J. Pesticide Sci.* **25**, 31–36 (2000).
62. Lew, R. R. Turgor and net ion flux responses to activation of the osmotic MAP kinase cascade by fludioxonil in the filamentous fungus *Neurospora crassa*. *Fungal Genet. Biol.* **47**, 721–726 (2010).
63. Colot, H. V. *et al.* A high-throughput gene knockout procedure for *Neurospora* reveals functions for multiple transcription factors. *Proc. Natl Acad. Sci. USA* **103**, 10352–10357 (2006).
64. Lew, R. R., Abbas, Z., Anderca, M. I. & Free, S. J. Phenotype of a mechanosensitive channel mutant, *mid-1*, in a filamentous fungus, *Neurospora crassa*. *Eukaryot. Cell* **7**, 647–655 (2008).
65. Lew, R. R., Kapishon, V. Ptk2 contributes to osmoadaptation in the filamentous fungus *Neurospora crassa*. *Fungal Genet. Biol.* **46**, 949–955 (2009).
66. Kaminsky, S. G. W., Garrill, A. & Heath, I. B. The relation between turgor and tip growth in *Saprolegnia ferax*: turgor is necessary, but not sufficient to explain apical extension rates. *Exp. Mycol.* **16**, 64–75 (1992).
67. Harold, R. L., Money, N. P. & Harold, F. M. Growth and morphogenesis in *Saprolegnia ferax*: is turgor required? *Protoplasma* **191**, 105–114 (1996).
68. Walker, S. K., Chitchoian, K., Yu, Y.-P., Christenhusz, G. M. & Garrill, A. Invasive hyphal growth: an F-actin depleted zone is associated with invasive hyphae of the oomycetes *Achlya bisexualis* and *Phytophthora cinnamomi*. *Fungal Genet. Biol.* **43**, 357–365 (2006).
69. Emerson, S. Slime, a plasmodial variant of *Neurospora crassa*. *Genetica* **34**, 162–182 (1963).
70. Perkins, D. D., Radford, A., Newmeyer, D. & Björkman, M. Chromosomal loci of *Neurospora crassa*. *Microbiol. Rev.* **46**, 426–570 (1982).
71. Leal-Morales, C. A. & Ruiz-Herrera, J. Alterations in the biosynthesis of chitin and glucan in the slime mutant of *Neurospora crassa*. *Exp. Mycol.* **9**, 28–38 (1985).
72. Bartnicki-Garcia, S., Bracker, C. E., Lippman, E. & Ruiz-Herrera, J. Chitosomes from the wall-less “slime” mutant of *Neurospora crassa*. *Arch. Microbiol.* **139**, 105–112 (1984).
73. Heath, I. B. & Steinberg, G. Mechanisms of hyphal tip growth: tube dwelling amebae revisited. *Fungal Genet. Biol.* **28**, 79–93 (1999).
- An alternative to a simple version of turgor-driven growth, emphasizing the role of the protoplast within the cell wall.**
74. Lew, R. R. Mass flow and pressure-driven hyphal extension in *Neurospora crassa*. *Microbiology* **151**, 2685–2692 (2005).
- The direct verification of mass flow in hyphal networks, and details of the magnitude of pressure gradients that are required to cause mass flow in a low-Reynolds-number environment.**
75. Robertson, N. F. & Rizvi, S. R. H. Some observations on the water-relations of the hyphae of *Neurospora crassa*. *Ann. Bot.* **32**, 279–291 (1968).
76. Money, N. P. Measurement of hyphal turgor. *Exp. Mycol.* **14**, 416–425 (1990).
- Measurement of turgor of the oomycete *S. ferax* using a variety of techniques. The methods would be the same for fungi. This article gives a good introduction to the thermodynamics of water potentials, experimental methodologies and experimental pitfalls.**
77. Ng, S. K., Liu, F., Lai, J., Low, W. & Jedd, G. A tether for Woronin body inheritance is associated with evolutionary variation in organelle positioning. *PLoS Genet.* **6**, e1000521 (2009).
78. Plamann, M. Cytoplasmic streaming in *Neurospora*: disperse the plug to increase the flow? *PLoS Genet.* **5**, e1000526 (2009).
79. Heaton, K. L. M., López, E., Maini, P. K., Fricker, M. D. & Jones, N. S. Growth-induced mass flows in fungal networks. *Proc. R. Soc. B* **277**, 3263–3274 (2010).
80. Sugden, K. E. P., Evans, M. R., Poon, W. C. K. & Read, N. D. Model of hyphal tip growth involving microtubule-based transport. *Phys. Rev. E Stat. Nonlin. Soft Matter Phys.* **75**, 031909 (2007).
81. Boudaoud, A. Growth of walled cells: from shells to vesicles. *Phys. Rev. Lett.* **91**, 018104 (2003).
82. Freitag, M., Hickey, P. C., Raju, N. B., Selker, E. U. & Read, N. D. GFP as a tool to analyze the organization, dynamics and function of nuclei and microtubules in *Neurospora crassa*. *Fungal Genet. Biol.* **41**, 897–910 (2004).
83. Nelson, G. *et al.* Calcium measurement in living filamentous fungi expressing codon-optimized aequorin. *Mol. Microbiol.* **52**, 1437–1450 (2004).

84. Hickey, P. C., Swift, S. M., Roca, M. G. & Read, N. D. Live-cell imaging of filamentous fungi using vital fluorescent dyes. *Methods Microbiol.* **34**, 63–87 (2005).
85. Virag, A. & Griffiths, A. J. A mutation in the *Neurospora crassa* actin gene results in multiple defects in tip growth and branching. *Fungal Genet. Biol.* **41**, 213–225 (2004).
86. Lee, I. H., Kumar, S. & Plamann, M. Null mutants of the *Neurospora* actin-related protein 1 pointed-end complex show distinct phenotypes. *Mol. Biol. Cell* **12**, 2195–2206 (2001).
87. Riquelme M., Roberson, R. W., McDaniel, D. P. & Bartnicki-Garcia S. The effects of *ropy-1* mutation on cytoplasmic organization and intracellular motility in mature hyphae of *Neurospora crassa*. *Fungal Genet. Biol.* **37**, 171–179 (2002).
88. Mourino-Perez, R. R., Roberson, R. W. & Bartnicki-Garcia, S. Microtubule dynamics and organization during hyphal growth and branching in *Neurospora crassa*. *Fungal Genet. Biol.* **43**, 389–400 (2006).
89. Ramos-Garcia, S. L., Roberson, R. W., Freitag, M., Bartnicki-Garcia, S. & Mourino-Perez, R. R. Cytoplasmic bulk flow propels nuclei in mature hyphae of *Neurospora crassa*. *Eukaryot. Cell* **8**, 1880–1890 (2009).
90. Roca, M. G., Kuo, H.-C., Lichius, A., Freitag, M. & Read, N. D. Nuclear dynamics, mitosis, and the cytoskeleton during the early stages of colony initiation in *Neurospora crassa*. *Eukaryot. Cell* **9**, 1171–1183 (2010).
91. Tucker, E. B. Cytoplasmic streaming does not drive intercellular passage in staminal hairs of *Setcreasea purpurea*. *Protoplasma* **137**, 140–144 (1987).
92. Sui, S. & Garrill, A. An F-actin-depleted zone is present at the hyphal tip of invasive hyphae of *Neurospora crassa*. *Protoplasma* **232**, 165–172 (2008).
93. Delgado-Álvarez, D. L. *et al.* Visualization of F-actin localization and dynamics with live cell markers in *Neurospora crassa*. *Fungal Genet. Biol.* **47**, 573–586 (2010).
94. Harris, S. D. *et al.* Polarisome meets Spitzenkörper: microscopy, genetics, and genomics converge. *Eukaryot. Cell* **4**, 225–229 (2005).
95. Gow, N. A. R. Transhyphal electrical currents in fungi. *J. Gen. Microbiol.* **130**, 3313–3318 (1984).
96. McGillviray, A. M. & Gow, N. A. R. The transhyphal electrical current of *Neurospora crassa* is carried principally by protons. *J. Gen. Microbiol.* **133**, 2875–2881 (1987).
97. Takeuchi Y., Schmid, J., Caldwell, J. H. & Harold, F. M. Transcellular ion currents and extension of *Neurospora crassa* hyphae. *J. Membr. Biol.* **101**, 33–41 (1988).
98. McGillviray, A. M. & Gow, N. A. R. Applied electrical fields polarize the growth of mycelial fungi. *J. Gen. Microbiol.* **132**, 2515–2525 (1986).
99. Lever, M. C. *et al.* pH and Ca^{2+} dependent galvanotropism of filamentous fungi: implications and mechanisms. *Mycol. Res.* **98**, 301–306 (1994).
100. Burstaller, W. Transport of small ions and molecules through the plasma membrane of filamentous fungi. *Curr. Rev. Microbiol.* **23**, 1–46 (1997).
101. Riquelme, M., Freitag, M., León-Hing, E. S. & Bowman, B. Live imaging of the secretory pathway in hyphae of *Neurospora crassa*. *Fungal Genet. Newsl.* **52** (Suppl.), 53 (2005).
102. Slayman, C. L. & Slayman, C. W. Measurement of membrane potentials in *Neurospora*. *Science* **136**, 876–877 (1962).
103. Potapova, T. V., Aslanidi, K. B., Belozerskaya, T. A. & Levina, N. N. Transcellular ionic currents studied by intracellular potential recordings in *Neurospora crassa* hyphae. Transfer of energy from proximal to apical cells. *FEBS Lett.* **241**, 173–176 (1988).
104. Zalokar, M. Growth and differentiation of *Neurospora crassa*. *Am. J. Bot.* **46**, 602–610 (1959).
105. Brody, J. P., Yager, P., Goldstein, R. E. & Austin, R. H. Biotechnology at low Reynolds number. *Biophys. J.* **71**, 3430–3441 (1996).
- Microfluidics at low Reynolds number is explored in detail from a physical viewpoint, pertinent to the nature of mass flow in hyphae.**
106. Berg, H. C. & Purcell, E. M. Physics of chemoreception. *Biophys. J.* **20**, 193–219 (1977).
107. Short, M. B. *et al.* Flows driven by flagella of multicellular organisms enhance long-range molecular transport. *Proc. Natl Acad. Sci. USA* **103**, 8315–8319 (2006).
108. Goldstein, R. E., Tuval, I. & van de Meent, J.-W. Microfluidics of cytoplasmic streaming and its implications for intracellular transport. *Proc. Natl Acad. Sci. USA* **105**, 3663–3667 (2008).
109. Van de Meent, J.-W., Tuval, I. & Goldstein, R. E. Nature's microfluidic transporter: rotational cytoplasmic streaming at high Péclet numbers. *Phys. Rev. Lett.* **101**, 178102 (2008).
110. Cosgrove, D. J. Analysis of the dynamic and steady-state responses of growth rate and turgor pressure to changes in cell parameters. *Plant Physiol.* **68**, 1439–1446 (1981).
- A detailed analysis of the nature of the interplay between turgor and cell expansion during cell growth. This analysis is pertinent to any walled cells that rely on turgor for growth.**
111. Bok, J. W. *et al.* Structure and function analysis of the calcium-related gene *spray* in *Neurospora crassa*. *Fungal Genet. Biol.* **32**, 145–158 (2001).
112. Sanders, D., Slayman, C. L. & Pall, M. L. Stoichiometry of H^+ /amino acid cotransport in *Neurospora crassa* revealed by current-voltage analysis. *Biochim. Biophys. Acta* **735**, 67–76 (1983).
113. Schloemer, R. H. & Garrett, R. H. Nitrate transport system in *Neurospora crassa*. *J. Bacteriol.* **118**, 259–269 (1974).
114. Blatt, M. R., Maurousset, L. & Meharg, A. A. High-affinity NO_3^-/H^+ cotransport in the fungus *Neurospora*: induction and control by pH and membrane voltage. *J. Membr. Biol.* **160**, 59–76 (1997).
115. Versaw, W. K. & Metznerberg, R. L. Repressible cation–phosphate symporter in *Neurospora crassa*. *Proc. Natl Acad. Sci. USA* **92**, 3884–3887 (1995).
116. Slayman, C. L. & Slayman, C. W. Depolarization of the plasma membrane of *Neurospora* during active transport of glucose: evidence for a proton-dependent cotransport system. *Proc. Natl Acad. Sci. USA* **71**, 1935–1939 (1974).

Competing interests statement

The author declares no competing financial interests.

Acknowledgments

The author thanks past and present members of the Lew laboratory, where research is funded by the Natural Sciences and Engineering Research Council of Canada.

FURTHER INFORMATION

Roger R. Lew's homepage: <http://www.yorku.ca/planters>

ALL LINKS ARE ACTIVE IN THE ONLINE PDF